WORKED SOLUTION

STEP III 2009, Question 9

Mechanics • Projectile motion

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Read the question before you go on. It is in the STEP Support Programme's question database:

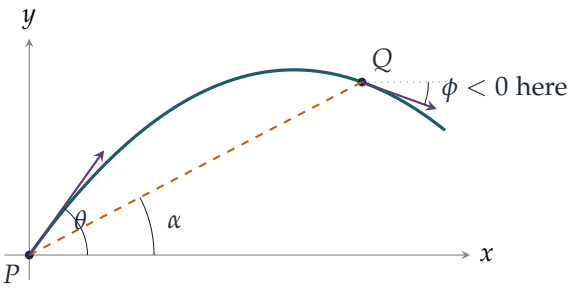
step.maths.org/questions/09-s3-q9

The key idea

In projectile motion the horizontal velocity never changes and the vertical velocity changes at a constant rate. So the average vertical velocity between P and Q is just the mean of its values at the two ends, and the gradient of the chord PQ is the mean of the gradients at the ends. That gives part (i) in one line. For part (ii), the trajectory equation is a quadratic in $\tan\theta$ (once we write $\sec^2\theta = 1 + \tan^2\theta$), and sums and products of roots do the rest.

Solution

We model the particle as moving freely under constant gravity g, with no air resistance. Take P as the origin, with the x-axis along the positive horizontal direction and the y-axis vertically up. Let u be the speed of projection and let $Q = (X, Y)$. We assume $X > 0$: Q lies on the positive side of P, so the horizontal velocity $u\cos\theta$ is positive and all the angles $\theta, \phi, \theta', \phi'$ lie in $(-\frac{\pi}{2}, \frac{\pi}{2})$. (If Q were vertically above or below P, the angles at P and Q would both be $\pm\frac{\pi}{2}$ and the question would say nothing.) Then $\tan\alpha = Y/X$.



Part (i)

The horizontal velocity is constant, $u\cos\theta$. The vertical velocity is $u\sin\theta$ at P, say v_Q at Q, and it decreases at the constant rate g, so it is a linear function of time. Hence, if the flight from P to Q takes time T,

$$X = (u\cos\theta)\,T, \qquad Y = \frac{u\sin\theta + v_Q}{2}\,T,$$

because the average of a linear function over an interval is the mean of its end values. At each point the direction of motion is the direction of the velocity, so $\tan\theta = \dfrac{u\sin\theta}{u\cos\theta}$ and $\tan\phi = \dfrac{v_Q}{u\cos\theta}$. Dividing,

$$\tan\alpha = \frac{Y}{X} = \frac{1}{2}\left(\frac{u\sin\theta}{u\cos\theta} + \frac{v_Q}{u\cos\theta}\right) = \tfrac{1}{2}(\tan\theta + \tan\phi),$$

which is $\tan\theta + \tan\phi = 2\tan\alpha$, as required.

Part (ii)

The quadratic. Eliminating t from $x = ut\cos\theta$, $y = ut\sin\theta - \frac{1}{2}gt^2$ gives the trajectory

$$y = x\tan\theta - \frac{gx^2}{2u^2\cos^2\theta} = x\tan\theta - \frac{gx^2}{2u^2}(1+\tan^2\theta).$$

Write $k = \dfrac{gX^2}{2u^2}$, which is positive. Since the path passes through (X, Y), $T = \tan\theta$ satisfies

$$kT^2 - XT + (Y + k) = 0.$$

The second trajectory has the same u, X and Y, so $\tan\theta'$ satisfies the same equation. The two trajectories are different, so $\theta \neq \theta'$, and since tan is one-to-one on $(-\frac{\pi}{2}, \frac{\pi}{2})$, $\tan\theta$ and $\tan\theta'$ are the *two* roots. Therefore

$$\tan\theta + \tan\theta' = \frac{X}{k}, \qquad \tan\theta\tan\theta' = \frac{Y+k}{k} = 1 + \frac{Y}{k}.$$

The first result. By the compound-angle formula,

$$\tan(\theta+\theta') = \frac{\tan\theta + \tan\theta'}{1 - \tan\theta\tan\theta'} = \frac{X/k}{-Y/k} = -\frac{X}{Y} = -\cot\alpha.$$

This needs $Y \neq 0$. If $Y = 0$ (so $\alpha = 0$ and $\cot\alpha$ is undefined), the product of the roots is 1, which says $\theta + \theta' = \frac{\pi}{2}$, the familiar pair of complementary angles for the same range on level ground. In that sense the result still holds.

The second result. Write $t = \tan\alpha = Y/X$ and $s = \tan\theta + \tan\theta' = X/k$. Then $Y/k = ts$, so $\tan\theta\tan\theta' = 1 + ts$. By part (i), applied to each trajectory,

$$\tan\phi = 2t - \tan\theta, \qquad \tan\phi' = 2t - \tan\theta'.$$

So

$$\tan\phi + \tan\phi' = 4t - s,$$
$$1 - \tan\phi\tan\phi' = 1 - (4t^2 - 2ts + \tan\theta\tan\theta') = 1 - 4t^2 + 2ts - 1 - ts = t(s - 4t).$$

These can't both be zero: if $\tan\phi' = -\tan\phi$ then $\tan\phi\tan\phi' = -\tan^2\phi \le 0 \ne 1$. When $Y \ne 0$ (so $t \ne 0$) and $s \ne 4t$,

$$\tan(\phi + \phi') = \frac{4t - s}{t(s - 4t)} = -\frac{1}{t} = -\cot\alpha = \tan(\theta + \theta').$$

In the other cases both $1 - \tan\phi\tan\phi'$ and $1 - \tan\theta\tan\theta'$ are zero ($t = 0$), or both $\tan(\phi + \phi')$ and $\tan(\theta + \theta')$ are infinite, so in every case $\phi + \phi'$ and $\theta + \theta'$ have the same tangent or are both odd multiples of $\frac{\pi}{2}$. Either way,

$$\theta + \theta' - (\phi + \phi') = n\pi \quad \text{for some integer } n.$$

Ask yourself Equal tangents only pin the angle down up to a multiple of π. How do we know the multiple is exactly one π?

We use the sizes of the angles. On each trajectory the gradient is $\tan\theta - \dfrac{gx}{u^2\cos^2\theta}$, which strictly decreases as x increases. Since $X > 0$, we get $\tan\phi < \tan\theta$, so $\phi < \theta$; likewise $\phi' < \theta'$. All four angles lie in $(-\frac{\pi}{2}, \frac{\pi}{2})$, so $0 < \theta - \phi < \pi$ and $0 < \theta' - \phi' < \pi$. Adding,

$$0 < n\pi = (\theta + \theta') - (\phi + \phi') < 2\pi,$$

so $n = 1$, and $\theta + \theta' = \pi + \phi + \phi'$, as required.

Checks

- Take $X = 1$, $Y = \frac{1}{2}$ and $u^2 = 4g$, so $k = \frac{1}{8}$. The quadratic becomes $T^2 - 8T + 5 = 0$, with roots $4 \pm \sqrt{11}$. Then $\tan(\theta + \theta') = 8/(1 - 5) = -2 = -\cot\alpha$. Numerically, $\theta + \theta' \approx 2.0344$ and $\pi + \phi + \phi' \approx 2.0344$, where $\tan\phi = 1 - \tan\theta$ and $\tan\phi' = 1 - \tan\theta'$.
- Level ground ($Y = 0$): $\theta + \theta' = \frac{\pi}{2}$, and by symmetry $\phi = -\theta$, $\phi' = -\theta'$. So $\pi + \phi + \phi' = \pi - \frac{\pi}{2} = \frac{\pi}{2}$, as it should be.
- We stepped the motion numerically to confirm part (i), and tested both results of part (ii) on 20,000 random choices of u, X and Y that admit two trajectories.

Common slips

- Concluding $\theta + \theta' = \pi + \phi + \phi'$ from equal tangents alone. Equal tangents give the angles only up to a multiple of π, and you need the inequalities $\phi < \theta$ and $\phi' < \theta'$ to settle which multiple.
- Forgetting to say why $\tan\theta$ and $\tan\theta'$ are the *two different* roots of the same quadratic. They are different because the trajectories are, and tan is one-to-one on $(-\frac{\pi}{2}, \frac{\pi}{2})$.

Beyond STEP

Part (ii) has a pleasant reading in terms of reversibility. Run either path backwards and you get a projectile from Q to P launched at angle $\phi + \pi$. So the four angles are tied together by a symmetry between the two ends, and the relation $\theta + \theta' = \pi + \phi + \phi'$ expresses it. Time-reversal symmetry of Newton's laws (without friction or drag) is a theme that runs right through university mechanics. For a thorough first-year treatment of projectiles and much more, see Kleppner and Kolenkow, *An Introduction to Mechanics*.

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